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Original Article

Reproductive genetics

Profiling mitochondrial DNA variant segregation during human preimplantation development: a prerequisite to preimplantation genetic testing for mitochondrial DNA-related disorders

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ABSTRACT

STUDY QUESTION: Is preimplantation genetic testing for mitochondrial DNA (mtDNA) disorders (PGT-mt) feasible at early compaction and blastocyst stages?

SUMMARY ANSWER: Pathogenic mtDNA variants segregate evenly among cell types and various lineages of a given embryo during preimplantation development, supporting the relevance of genetic analyses performed on Day 4 blastomere and on Day 5 or 6 trophectoderm (TE) samples.

WHAT IS KNOWN ALREADY: PGT-mt is validated at cleavage stage (Day 3 of development). However, its feasibility at later stages is questionable, as little is known regarding the segregation of pathogenic mtDNA variants during preimplantation development. Since mtDNA replication is silenced until the blastocyst stage (Day 5 or 6), uneven mtDNA segregation between preimplantation embryo cellular lineages known as a 'bottleneck' effect, cannot be excluded, posing a challenge for PGT-mt.

STUDY DESIGN, SIZE, DURATION: We analyzed 112 'mito' embryos carrying pathogenic mtDNA variants and 28 control embryos with mtDNA polymorphism. Heteroplasmy levels were assessed in single cells of the TE, in different parts of blastocysts (inner cell mass and TE), and at three time points of development, namely cleavage (Day 3), early compaction (Day 4), and blastocyst stages (Day 5 or 6).

PARTICIPANTS/MATERIALS, SETTING, METHODS: As part of clinical PGT, a blastomere biopsy was performed at cleavage or early compaction stages (Day 3 or 4) on 112 'mito' and 21/28 control embryos. Further analysis was carried out at Day 5 or 6 on 51 embryos deemed unsuitable for uterine transfer and donated to research. Heteroplasmy levels were determined by semi-quantitative PCR amplification of (i) the mtDNA pathogenic variants with additional enzymatic digestion or (ii) the mtDNA polymorphic hypervariable region 2.

MAIN RESULTS AND THE ROLE OF CHANCE: Here, we first show that mtDNA variants segregate evenly among blastomeres during early compaction (Day 4), supporting the feasibility of PGT-mt at this stage. We also found that mtDNA ratios remain stable between cleavage and blastocyst stages. Yet, the substantial variation of heteroplasmy levels occurring among single TE cells in 1/8 embryos suggests that PGT is only feasible when at least 5–10 cells are collected by standard TE biopsy.

LIMITATIONS, REASONS FOR CAUTION: This study sheds light on mtDNA segregation in human preimplantation embryo development. Its limitation lies in the scarcity of the material and the small number of embryos carrying a specific pathogenic mtDNA variant. Furthermore, the study of single cells from TE was performed on control embryos only.

WIDER IMPLICATIONS OF THE FINDINGS: By supporting the relevance of blastocyst biopsy in the context of PGT for pathogenic mtDNA variants, this study contributes to the general trend of postponing the biopsy to later stages of embryonic development. However, particular attention should be paid to the number of TE cells tested. Due to the potential variation of mutant load during *in utero* development, a control amniocentesis for evolutive pregnancies following the transfer of heteroplasmic embryos is still recommended.

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Introduction

Mitochondrial DNA (mtDNA) diseases are severe, maternally inherited, multisystem disorders presenting variable and overlapping clinical characteristics (Chinnery and Schon, 2003; Gorman *et al.*, 2016). This clinical heterogeneity may be ascribed at least in part to heteroplasmy, i.e. the coexistence of two or more mtDNA molecules, such as mutant and wild-type mtDNA in the same cell, a feature known to vary with time and among tissues (Sondheimer *et al.*, 2011; Parakatselaki and Ladoukakis, 2021; Glynos *et al.*, 2023).

In an attempt to avoid transmission of these debilitating conditions to their progeny, carrier mothers often request preimplantation genetic testing (PGT-mt) and prenatal diagnosis (Steffann *et al.*, 2021). These antenatal diagnostic procedures are based on the assessment of pathogenic mtDNA variant loads (mutant load). Yet, these procedures are challenging, and three major considerations must be addressed: (i) the definition of variant-specific thresholds under which the risk of severe clinical expression is expected to be low; (ii) the whole-body representativity of the mutant load assessed on a single tissue/cell type; and (iii) the possible variability of the mutant load throughout pre and postnatal development.

Assessment of pathogenic mtDNA variant load on one to two blastomeres at cleavage stage (Day 3) has been previously shown to be representative of the mutant load in the remaining cells (Steffann *et al.*, 2014; Sallevelt *et al.*, 2017). Biopsy at later stages of preimplantation development seems to have lesser detrimental effects on embryonic development (Scott *et al.*, 2013; Bar-El *et al.*, 2016; Coll *et al.*, 2018; Orvieto *et al.*, 2020), however, little is known regarding the relevance of blastomere biopsies at the early compaction stage (Day 4) and trophectoderm (TE) biopsies at blastocyst stage (Day 5 or 6) for PGT-mt (Treff *et al.*, 2012; Heindryckx *et al.*, 2014; Ji *et al.*, 2021; Spath *et al.*, 2021). Even if mutant loads seem to remain stable between blastomeres at Day 3 and embryos analyzed at Day 5 or 6 (Monnot *et al.*, 2011; Steffann *et al.*, 2015; Sallevelt *et al.*, 2017; Janssen *et al.*, 2024), Mitalipov *et al.* reported a case of a child who developed a complex multisystem disorder born after PGT-mt at the blastocyst stage for the m.3243A>G pathogenic variant of the mtDNA associated with mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes syndrome. In this case, TE biopsy showed a low mutant load (12%), but a nearly 50% mutant load was found in postnatal tissues (Mitalipov *et al.*, 2014). Discordance between the two mutant loads could be ascribed to a difference in mutant loads between the TE (which will eventually become the placenta and extra-embryonary tissues) and the inner cell mass (ICM, which will become the embryo itself). A skewed mtDNA segregation may indeed result from a 'bottleneck' phenomenon at this early stage of embryonic development, potentially caused by the arrest of mtDNA replication in early preimplantation embryos. As the embryonic cells continue to divide, the arrest of mtDNA replication results in the reduction of the number of mtDNA molecules per cell and, consequently, in potential rapid shifts of heteroplasmy (Floros *et al.*, 2018). The exact time of this putative 'bottleneck' is presently unknown, but it has a major impact on the ideal timing of PGT biopsies in the context of mtDNA disorders. Provided that blastocyst TE is representative of the ICM, it would indeed be tempting to implement a less aggressive TE biopsy procedure for safer and less deleterious PGT-mt.

To explore the feasibility of PGT-mt for carrier women at the late stages of preimplantation embryonic development, we traced mtDNA segregation in embryos carrying known pathogenic mtDNA variants and in controls at different stages of embryonic development, from cleavage (Day 3), early compaction (Day 4), and blastocyst (Day 5 or 6) stages.

Materials and methods

A total of 33 couples agreed to donate their affected embryos to research and provided written informed consent. The French Biomedical Agency approved this study according to the national health legislation (authorization: 16 April 2016/renewal: 26 October 2021 for 5 years).

Each of the 136 embryos included in this study was sampled at least twice (Supplementary Table S1 and Fig. S1). The samples were taken simultaneously (either two different blastomeres or TE and ICM) and/or sequentially (the same embryo was sampled at different developmental stages). A total of 112 embryos carrying pathogenic mtDNA variants composed the mito group (Supplementary Fig. S1). Heteroplasmy was determined in the 28 embryos of the control group by assessing the hypervariable region 2 (HV2) of mtDNA. The HV2 region is located between nucleotides 303 and 315 in the mtDNA displacement loop (D loop). It is composed of a highly polymorphic homopolymeric tract of cytosines widely used for mtDNA segregation analyses (Marchington *et al.*, 1997; Lee *et al.*, 2004; Steffann *et al.*, 2006). Of these 28 control embryos, 24 carried a pathogenic nuclear DNA variant diagnosed by conventional PGT and were, therefore, unsuitable for uterine transfer. Four embryos belonging to the mito group (embryo numbers 46–49; Supplementary Table S1) were also heteroplasmic for HV2. Consequently, these embryos were included in both the mito and the control group, bringing the control group to a total of 28 embryos.

Of the 28 embryos in the control group, 21 were biopsied either at Day 3 (n = 28 cells) or at Day 4 (n = 10 cells) for clinical PGT, and the remaining PCR amplification products of the blastomeres were reamplified for the HV2 region. Among the 25 out of 28 control embryos that reached the blastocyst stage, 21 embryos had their TE sampled, and 7 embryos had ICM samples collected when the ICM could be clearly identified. After TE and/or ICM sampling, the remaining cells were analyzed as one sample and termed the 'rest' (n = 19). Additionally, 43 single cells were dissociated from TE samples of 8 control embryos using Trypsine or Accutase as described (Lammers *et al.*, 2023) and analyzed separately.

A total of 97 'mito' embryos were biopsied at cleavage stage (Day 3) in the context of PGT, and 190 blastomeres were analyzed (Supplementary Fig. S1). Among them, 31 embryos were at risk of carrying the m.3243A>G variant of the *MT-TL1* gene (60 blastomeres); 36 embryos carried the m.8344A>G variant of the *MT-TK* gene (71 blastomeres); 18 carried the m.8993T>C or the m.8993T>G variants (36 blastomeres); and 5 the m.9185T>C variant of the *MT-ATP6* gene (10 blastomeres); and 7 carried the m.13094T>C variant of the *MT-ND5* gene (13 blastomeres). Additionally, 2 or 3 blastomeres were obtained from 15 embryos at the early compaction stage (Day 4) for a total of 31 blastomeres (1 embryo: m.3243A>G; 3 embryos: m.8344A>G, 7 embryos: m.8993T>C, 1 embryo: m.8993T>G, and 3 embryos: m.9185T>C). Among the 28/112 'mito' embryos that reached the blastocyst stage, 16 embryos were collected as a whole (7 m.3243A>G, 1 m.8993T>C, and 8 m.8344A>G), 2 'rest' (1 m.8993T>C and 1 m.8344A>G), as well as 12 TE (1 m.3243A>G; 4 m.8344A>G; 1 m.8993T>C, 4 m.8993T>G; and 2 m.13094T>C), and 3 ICM samples (3 m.8344A>G).

All embryo biopsies were performed as described in previous studies (Xu *et al.*, 2014; Aoyama and Kato, 2020; Kai *et al.*, 2022). Samples were collected in an IVF laboratory using a stripper-

controlled fine heat-polished glass micropipette in drops of PBS (Sigma Aldrich, Lyon, France) supplemented with 0.1% polyvinyl alcohol (Sigma Aldrich) and deposited in 3 µl of lysis buffer (200 mM KOH, 50 mM DTT) under visual control using an inverted microscope. PCR amplification and sample handling were performed in a laboratory certified for PGT procedures. Assessment of HV2 heteroplasmy levels was performed by fluorescent PCR, as previously described (Lee *et al.*, 2004) (Supplementary Fig. S2). For heteroplasmic samples containing multiple HV2 alleles, the percentage of the prevalent HV2 allele was calculated and considered as the 'heteroplasmy level'. The mutant mtDNA load was assessed by semi-quantitative fluorescent PCR followed by enzymatic restriction digestion as previously described (Gigarel *et al.*, 2005; Monnot *et al.*, 2011, 2013) (Supplementary Fig. S3). The reproducibility of this type of assay reportedly shows a variability of up to 2% (Gigarel *et al.*, 2005).

Results

Among the 28 control embryos tested, 10/28 were homoplasmic for one HV2 allele and all the samples were concordant (14 blastomeres, 10 TE samples, 3 ICM samples, and 9 'rest'). The remaining 18 control embryos were heteroplasmic for at least two HV2 alleles and considered informative (Supplementary Fig. S1). Among them, four embryos (Nos 46–49; Supplementary Table S1) were carriers of mtDNA mutation and were also included in the mito group.

A high correlation of HV2 heteroplasmy levels was observed between the two blastomeres collected from 7 informative embryos at Day 3 ($R^2 = 0.94$; $P < 0.001$) with an average difference in HV2 heteroplasmy of 2% (range: 0–9%; median = 1%; SD = 3.1) (Fig. 1A). Similarly, a concordant mutant load was found between two blastomeres collected from 93/97 'mito' embryos at Day 3 ($R^2 = 0.99$; $P < 0.001$) (Fig. 1A). The average difference in mutant loads between blastomeres was 0.6% (range: 0–13%; median = 0%; SD = 3.0). This high correlation rate was found for all variants analyzed, regardless of pathogenic variant or mutant load. Only 14/93 embryos of the mito group showed discrepant mutant loads ranging from 5% to 13% between two blastomeres at Day 3, which exceeds the expected differences in this type of PCR assay. Among these 14 embryos, almost all (n = 12) carried a small difference of <10% of the mutant load between the two blastomeres. The two remaining embryos (Nos 78 and 81, Supplementary Table S1) carried the m.8344A>G variant of the *MT-TK* gene, and the blastomeres displayed 26 and 39, and 79 and 90 mutant loads, respectively.

At Day 4, we found that HV2 heteroplasmy rate and mutant mtDNA load were comparable between two or three blastomeres obtained from 3 control and 15 'mito' embryos ($R^2 = 0.99$; $P < 0.001$) (Fig. 1B). The average difference in HV2 heteroplasmy rate was only 2% (range: 0–8%; SD = 4.6; n = 3 blastomere pairs) for the control group, as well as for the mutant load in the mito group (range: 0–6%; SD = 1.6; n = 17 blastomere pairs) (Fig. 1B).

Segregation of mtDNA at the blastocyst stage was also investigated by comparing mutant loads (n = 5 embryos) and HV2 heteroplasmy levels (n = 10 embryos) in different tissues: the ICM (that will develop into the future embryo, n = 7), the TE (that is biopsied in PGT procedures, n = 17), and the 'rest' (mix of TE and ICM, n = 12 samples) (Fig. 2). Again, strong correlations in mutant load and HV2 heteroplasmy rate were found between 7 pairs of TE and ICM ($R^2 = 0.93$; $P < 0.001$), and between 14 pairs of TE and 'rest' ($R^2 = 0.97$; $P < 0.001$) (Fig. 2). Additionally, mtDNA populations were investigated in 43 single cells isolated from TE samples from 8 control blastocyst embryos (Day 5 or 6). In 7/8 embryos, a consistent distribution of the HV2 molecules was found among the different cells, with differences ranging between 0% and 10% (average SD = 1.7) (Fig. 3). Notably, 1/8 embryos (embryo number 122; Supplementary Table S1) presented a marked discrepancy in heteroplasmy rate (up to 18% difference between cells, SD = 6.3; n = 5 cells). Interestingly, mutation loads assessed on 4 TE biopsies were similar to the average loads of the corresponding single cells, suggesting that a small number of cells (from 3 to 8) is sufficient to predict the TE heteroplasmy levels (Fig. 3).

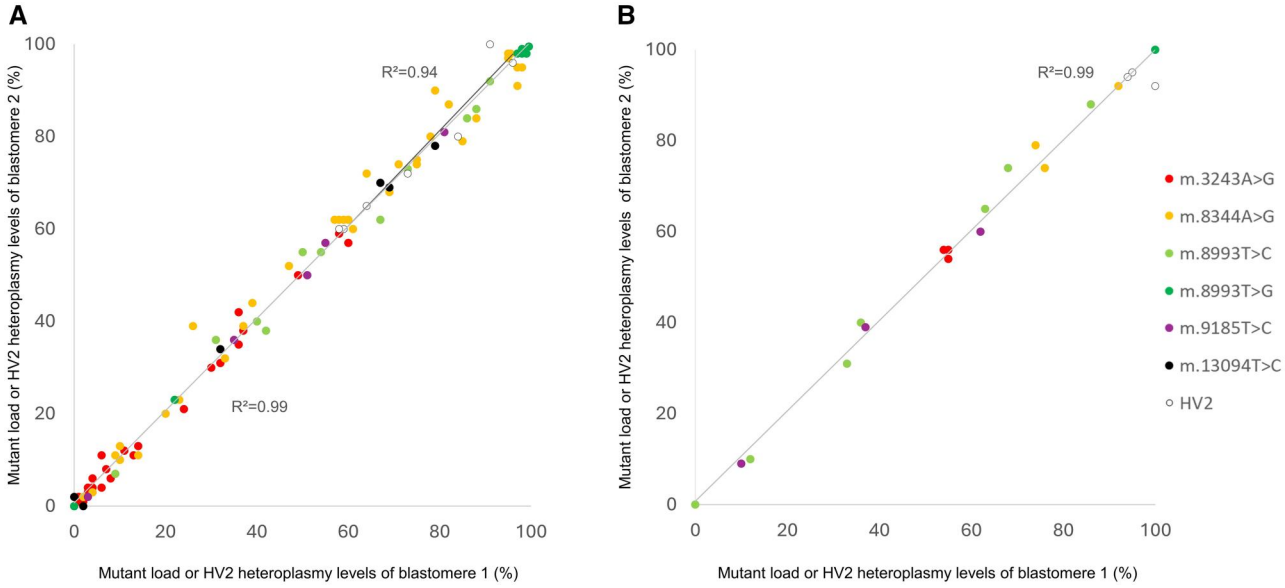


Figure 1. Intercellular stability of heteroplasmy in cleavage stage (Day 3) and early compaction (Day 4) embryos. Scatter plots examining the relationship between mutant load or hypervariable region 2 (HV2) heteroplasmy levels in two distinct blastomeres collected from the same embryo at (**A**) Day 3 and (**B**) Day 4 of preimplantation development. Linear regression showed a high correlation between the 2 blastomeres for (A) mutant load (gray-fitted line; $R^2 = 0.99$, $P < 0.001$) and HV2 heteroplasmy levels (black-fitted line; $R^2 = 0.94$, $P < 0.001$) at Day 3 and (B) mutant load and HV2 heteroplasmy levels (gray-fitted line; $R^2 = 0.99$, $P < 0.001$) at Day 4.

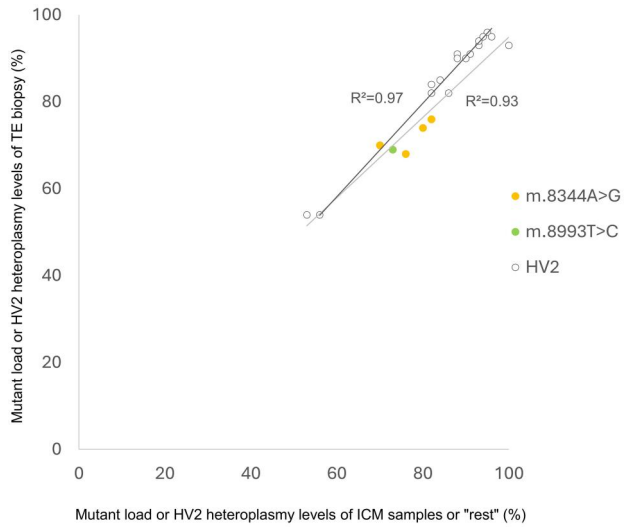


Figure 2. Inter-tissue stability of heteroplasmy in the blastocyst embryo. Scatter plot examining the relationship between mutant load or hypervariable region 2 (HV2) heteroplasmy levels in two distinct tissues sampled at the same point in time from the same blastocyst (Day 5 or 6), i.e. inner cell mass (ICM), trophectoderm (TE) biopsy, or the remaining embryonic cells after ICM and/or TE sampling (rest). Linear regression showed a high correlation between TE and ICM samples (gray-fitted line; $R^2 = 0.93$, $P < 0.001$) and between TE and 'rest' samples (black-fitted line; $R^2 = 0.97$, $P < 0.001$).

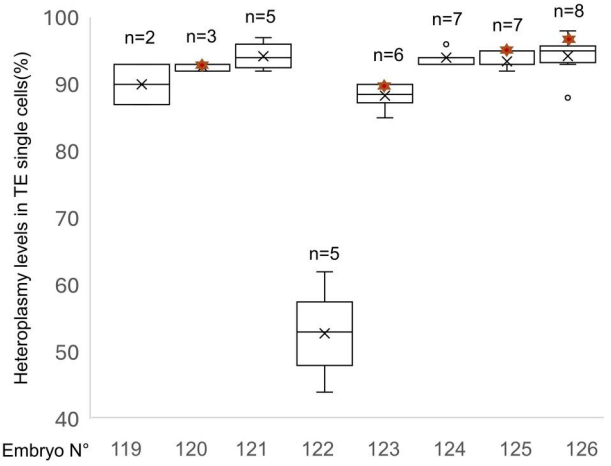


Figure 3. Intercellular variability of heteroplasmy in blastocyst embryos at Day 5 or 6 of development. Hypervariable region 2 (HV2) heteroplasmy levels were assessed in single cells dissociated from trophectoderm (TE) samples originating from 8 distinct control embryos. The number of cells analyzed is indicated as 'n'. Box plots represent the median, the first, and third quartiles of the single-cell HV2 heteroplasmy levels in each embryo. The red star represents the HV2 heteroplasmy levels found in the TE biopsy containing approximately 5–10 cells of the corresponding blastocyst.

To search for possible shifts in segregation of mtDNA molecules over time, we investigated HV2 heteroplasmy rate (n = 11 embryos) or mutant mtDNA load (n = 12 embryos) at cleavage and blastocyst stages. A strong correlation of HV2 heteroplasmy rate was found between 19 blastomeres biopsied at Days 3 or 4 and their respective 13 TE samples at Days 5 or 6 ($R^2 = 0.93$; $P < 0.001$) (Fig. 4A) as well as between 15 blastomeres and the 9 respective 'rest' ($R^2 = 0.86$; $P < 0.001$) (Fig. 4B). A similar correlation of the mutant mtDNA load was found between 22 blastomeres and their respective 12 TE samples ($R^2 = 0.96$; $P < 0.001$) (Fig. 4A) as well as between 34 blastomeres and their respective 18 'rest' or whole embryo ($R^2 = 0.95$; $P < 0.001$) (Fig. 4B). This stability was observed for all mtDNA variants analyzed.

Discussion

Profiling the segregation of pathogenic and non-pathogenic mtDNA variants in a series of 136 preimplantation human embryos, we observed a high degree of concordance of mutant mtDNA load and HV2 heteroplasmy levels between blastomeres at Day 3 of development. This stability gives additional support to the previous results reported by our group (Steffann *et al.*, 2014) and others (Sallevelt *et al.*, 2017), even if a marked difference of over 10% between blastomere pairs was observed in 2 embryos at Day 3, all carriers of the m.8344A>G variant of the *MT-TK* gene. Remarkably, none of these 2 discordant embryos were considered for uterine transfer, their mutant loads largely exceeding the tolerance threshold of 18%, as proposed by Hellebrekers *et al.* (2012). Moreover, we first show here that the mtDNA variants segregated evenly among blastomeres during the early compaction stage (Day 4), regardless of the mtDNA variant or heteroplasmy levels. Overall, our data indicate that wild-type and pathogenic mtDNA variants are homogeneously distributed among different cells of a given embryo at cleavage (Day 3) and early compaction stages (Day 4), allowing for feasible PGT based on one to two cells at these stages of development.

Consistently, our data show that pathogenic and non-pathogenic mtDNA variants segregated evenly among blastocyst compartments (TE and ICM), suggesting that TE biopsy at the blastocyst stage is representative of the ICM, which will further develop into the embryo. Furthermore, in 8/9 embryos analyzed, the HV2 heteroplasmy levels were stable among TE single cells. A marked difference in HV2 heteroplasmy levels between single cells was observed in only one embryo, carrier of an intermediate HV2 heteroplasmy level of around 50%. It is worth noting that such marked variations were not found in embryos with a highly homogeneous mtDNA population, suggesting that significant variations are not expected in embryos carrying low mtDNA mutant loads that are eligible for uterine transfer. Unfortunately, we were unable to confirm this hypothesis since blastocysts with low mutant loads were transferred in the context of clinical PGT. The significant intercellular variations reported here might result from a 'bottleneck' effect, particularly at this stage, when the number of mtDNA molecules is expected to be low before mtDNA replication resumes. Such variations might also explain the geographic variations in mutant load found in some placentas, which is a tissue derived from the TE (Vachin *et al.*, 2018). Finally, even if HV2 heteroplasmy analysis in TE samples suggests the feasibility of PGT-mt at the blastocyst stage, the presence of a mtDNA pathogenic variant can affect mtDNA segregation differently and warrants further studies.

Lastly, we have shown that mutant/wild-type mtDNA ratios remained stable between cleavage or early compaction and blastocyst stages, in 28 embryos tested, irrespective of their mutant load and mtDNA pathogenic variant.

Trophectoderm biopsy at Day 5 or 6 blastocyst stage is an increasingly preferred technique for nuclear DNA-related PGT, as it is considered less susceptible to damage the embryo and may be useful in the laboratory setup (Kokkali *et al.*, 2007; De Rycke *et al.*, 2017; Spinella *et al.*, 2023; Arroja *et al.*, 2024). However, before shifting to TE biopsy for mtDNA disorders, more data on mtDNA segregation in human blastocysts are needed. Indeed, if mtDNA segregation differs in extra-embryonary tissues, extension of PGT

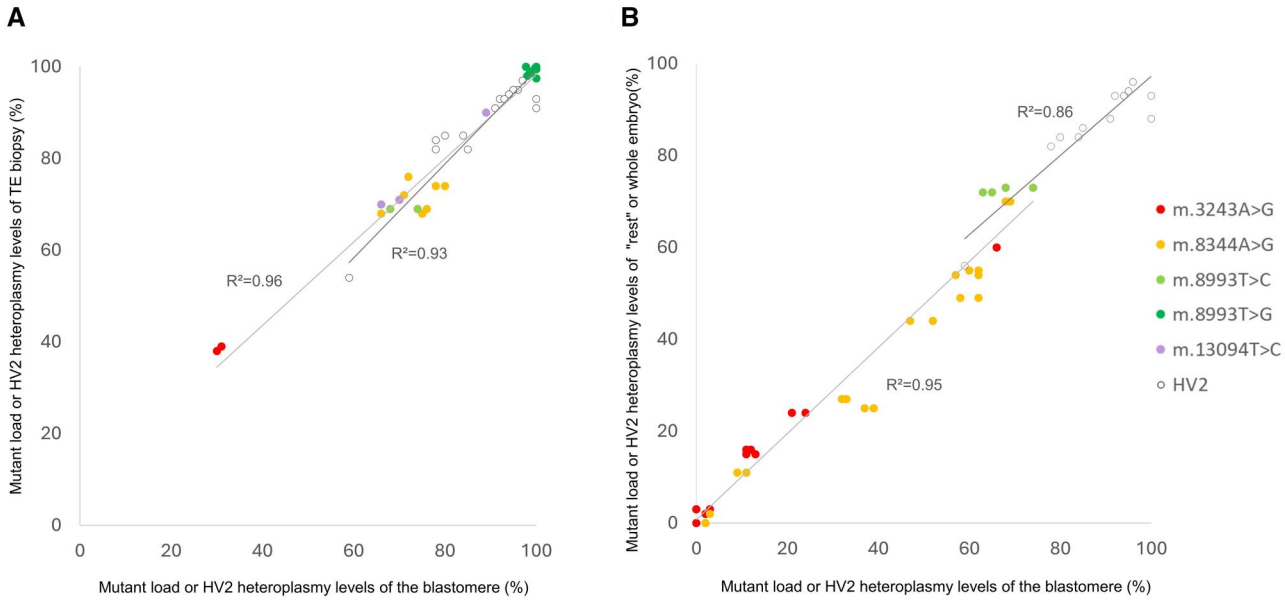


Figure 4. Stability of heteroplasmy levels within time in the developing preimplantation embryo. Scatter plots examining the relationship between mutant load or hypervariable region 2 (HV2) heteroplasmy levels at Day 3 or 4 on single blastomeres and at Day 5 or 6 from the corresponding blastocysts, either by (**A**) trophectoderm (TE) biopsy, or on (**B**) blastocyst after TE sampling; (rest) or whole embryo. Linear regression showed a high correlation between (A) blastomeres and TE biopsy for mutant load (gray-fitted line; $R^2 = 0.96$, $P < 0.001$) and for HV2 heteroplasmy levels (black-fitted line; $R^2 = 0.93$, $P < 0.001$) and between (B) blastomeres and 'rest' or whole embryo for mutant load (gray-fitted line; $R^2 = 0.95$, $P < 0.001$) and for HV2 heteroplasmy levels (black-fitted line; $R^2 = 0.86$, $P < 0.001$).

to TE biopsies in mtDNA-related disorders would be compromised. Here, we show that the assessment of heteroplasmy levels on a classical TE biopsy of sufficient size (5–10 cells) overcomes the potential intercellular variations and, therefore, would be representative of the blastocyst.

In conclusion, our study supports the notion of random segregation of mtDNA molecules during preimplantation human development. However, because some shifts can still occur at later stages of prenatal development and owing to the small number of samples available, a prenatal diagnosis on amniotic fluid is still recommended when a pregnancy is obtained after the transfer of an embryo carrying a heteroplasmic pathogenic mtDNA variant at the preimplantation stage. Furthermore, to definitely establish the reliability of PGT-mt, postnatal monitoring of the children born after PGT-mt is necessary. Future studies on larger cohorts will hopefully further contribute to a better understanding of mtDNA segregation in early human embryos and improve current PGT protocols for mtDNA-related disorders.

Supplementary data

Supplementary data are available at *Human Reproduction* online.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Authors' roles

Conceptualization, J.S.; Methodology J.S., N.G., P.R.; Investigation, P.R., A.Ma., K.C., N.G.; Writing-Original Draft, P.R. and J.S.; Writing-Review and Editing, A.Mu., N.F., A.Ma., P.R., and J.S.; Funding Acquisition, J.S., P.R., and K.C.; Resources J.S., S.M., A.R., N.F.; Supervision J.S.

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Conflict of interest

None declared.

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